

B. TECH
(SEM VI) THEORY EXAMINATION 2022-23
TRANSPORTATION ENGINEERING

*Time: 3 Hours**Total Marks: 70*

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief. **2 x 7 = 14**

- (a) Classify the road on the basis of location and function
- (b) Define camber and gradient
- (c) Explain space headway and time headway
- (d) Define traffic density and traffic volume.
- (e) Explain in brief about the composite pavement.
- (f) List and explain different type of temperature stresses
- (g) Explain the function of tack coat and seal coat

SECTION B

2. Attempt any three of the following: **7 x 3 = 21**

- (a) Classify the difference between all three-road plans implemented in India by considering any 4 parameters.
- (b) Find the stopping sight distances for level, ascending and descending grades on a road with a design speed of 100km/h. Assume total reaction time as 2.5 second and coefficient of friction as 0.36. Take gradient of a road as 1 in 40
- (c) Calculate the coefficient of friction, if:
 - (i) Breaking distance equal to 5.8 m speed of vehicle 30 kmph
 - (ii) Initial speed of vehicle 40 kmph duration of break application 1.8 second
 - (iii) Breaking distance 7 m and duration of break application is 1.4 second
- (d) Enlist tests which are performed for aggregates. Explain any one test. Calculate aggregate crushing value if weight of aggregate before and after the test is 500gms and 400gms respectively.
- (e) Describe the construction steps of bituminous carpeting.

SECTION C

3. Attempt any one part of the following: **7 x 1 = 7**

- (a) Explain the historical development of road construction.
- (b) Provide salient features of third twenty-year road plan.

4. Attempt any one part of the following: **7 x 1 = 7**

- (a) Calculate the length of transition curve using the following data:
 Design speed= 65kmph
 Radius of circular curve= 220m
 Pavement with extra widening= 7.5m
 Allowable rate of introduction of super elevation (pavement rotated about the center line) = 1 in 150

- (b) What is the extra widening required for a pavement of 7 m width on a horizontal curve of radius 200 m, if the longest wheel of vehicle expected on the road is 6.5 m and the design speed is 80 km/h?

5. Attempt any *one* part of the following:

7 x 1 = 7

- (a) Explain
- (i) 30th highest hourly traffic volume with neat graph.
 - (ii) 98th and 15th percentile speed with neat graph
- (b) Enlist and discuss briefly the various factors considered in the design of rotary intersection. Calculate the density of stream if average spacing between vehicles in a traffic stream is 60 m

6. Attempt any *one* part of the following:

7 x 1 = 7

- (a) Using the data given below calculate the wheel load stresses at a) interior and b) edge region of a cement concrete pavement using westergard's stress equation. Also determine the probable location where the crack is likely to develop due to corner loading.

Wheel load, $P = 5100 \text{ kg}$

Modulus of elasticity of cement concrete pavement, $E = 300000 \text{ kg/cm}^2$

Thickness, $h = 18 \text{ cm}$

Poisson's ratio of concrete = 0.15

Modulus of subgrade reaction, $K = 6 \text{ kg/cm}^3$

Radius of contact area, $a = 15 \text{ cm}$

- (b) The normal flow of traffic on crossroads A and B are 650 and 500 vehicles per hour respectively. The saturation of flow for roads A and B are estimated as 1500 and 1200 vehicles per hour respectively. The all-red time for pedestrians to cross is 12 seconds. Design a two-phase traffic system by Webster's method.

7. Attempt any *one* part of the following:

7 x 1 = 7

- (a) Provide the essential requirements of soil properties considered suitable for the construction of highways subgrade.
- (b) Explain the difference between Water Bound Macadam (WBM) and Wet mix macadam (WMM) in detail.